

DIGITAL INDIA AND INDIA STACK: UPI, AADHAAR

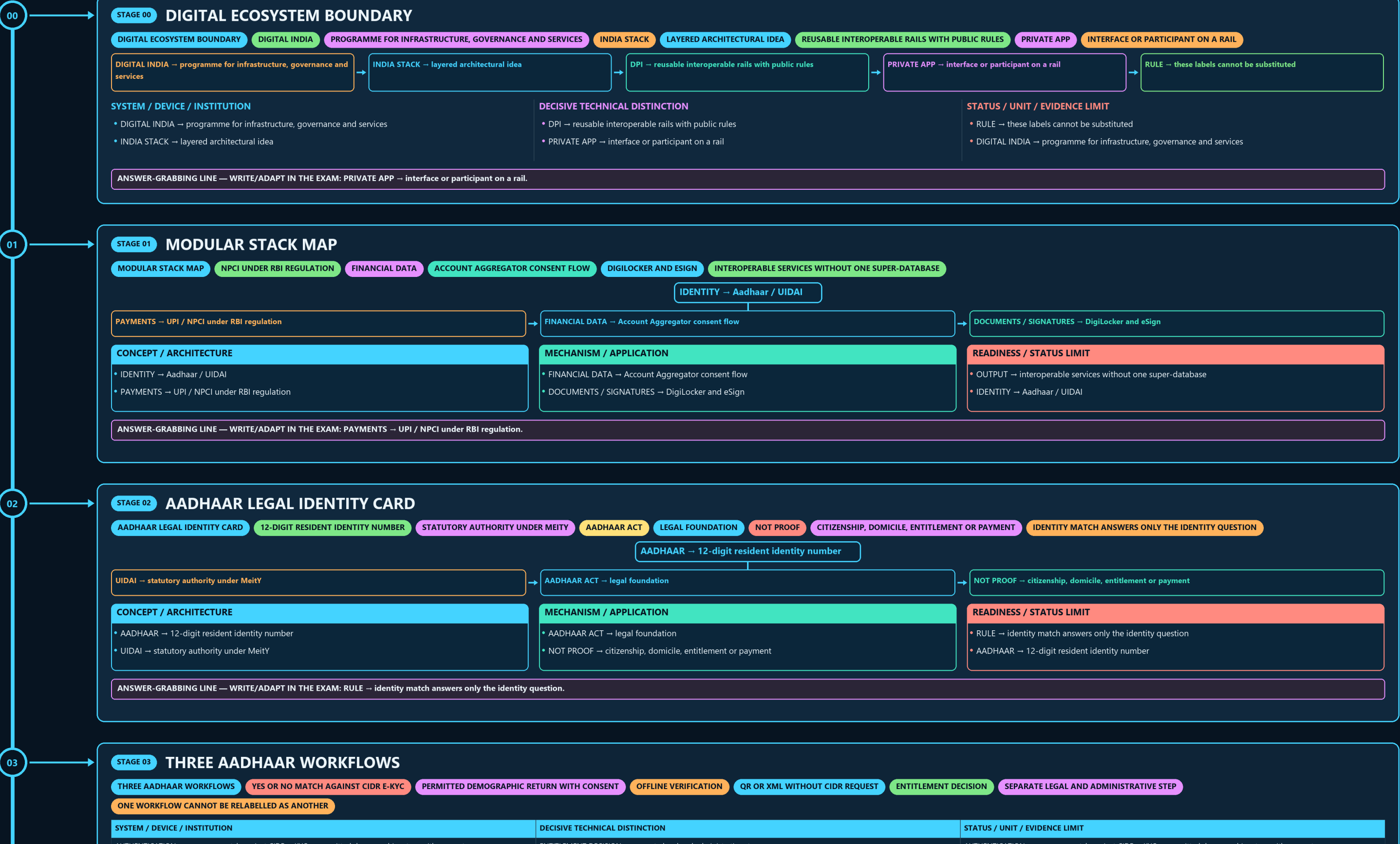
DIGITAL ECOSYSTEM BOUNDARY → MODULAR STACK MAP → AADHAAR LEGAL IDENTITY CARD → THREE AADHAAR WORKFLOWS → UPI TRANSACTION CHAIN → PAYMENT-RAIL MATRIX

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- AADHAAR → 12-digit resident identity number

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → identity match answers only the identity question.

STAGE 03 THREE AADHAAR WORKFLOWS

SEPARATE LEGAL AND ADMINISTRATIVE STEP

ONE WORKFLOW CANNOT BE RELABELLED AS ANOTHER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
AUTHENTICATION → yes or no match against CIDR e-KYC → permitted demographic return with consent	ENTITLEMENT DECISION → separate legal and administrative step	AUTHENTICATION → yes or no match against CIDR e-KYC → permitted demographic return with consent
OFFLINE VERIFICATION → QR or XML without CIDR request	TRAP → one workflow cannot be relabelled as another	OFFLINE VERIFICATION → QR or XML without CIDR request

SYSTEM / DEVICE / INSTITUTION

- AUTHENTICATION → yes or no match against CIDR e-KYC → permitted demographic return with consent
- OFFLINE VERIFICATION → QR or XML without CIDR request

DECISIVE TECHNICAL DISTINCTION

- ENTITLEMENT DECISION → separate legal and administrative step
- TRAP → one workflow cannot be relabelled as another

STATUS / UNIT / EVIDENCE LIMIT

- AUTHENTICATION → yes or no match against CIDR e-KYC → permitted demographic return with consent
- OFFLINE VERIFICATION → QR or XML without CIDR request

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → one workflow cannot be relabelled as another.

STAGE 04 UPI TRANSACTION CHAIN

PAYMENT-SYSTEM REGULATION



STARTING CONDITION / INPUT

- USER APP → initiates payment instruction
- UPI / NPCI → interoperable messaging and switching rail

SCIENTIFIC OR ENGINEERING MECHANISM

- BANK OR PERMITTED INSTRUMENT → funding source
- RBI FRAMEWORK → payment-system regulation

OUTPUT / FAILURE BOUNDARY

- RULE → rail, app, account and regulator are distinct
- USER APP → initiates payment instruction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE – rail, app, account and regulator are distinct.

STAGE 05 PAYMENT-RAIL MATRIX

DIGITAL PAYMENT DOES NOT MEAN UPI ALONE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
UPI → instant retail payment instruction	AePS → Aadhaar-authenticated correspondent banking	TRAP → digital payment does not mean UPI alone
IMPS / NEFT / RTGS → distinct interbank settlement modes	BBPS / NACH → bills and recurring bulk mandates	UPI → instant retail payment instruction

SYSTEM / DEVICE / INSTITUTION

- UPI → instant retail payment instruction
- IMPS / NEFT / RTGS → distinct interbank settlement modes

DECISIVE TECHNICAL DISTINCTION

- AePS → Aadhaar-authenticated correspondent banking
- BBPS / NACH → bills and recurring bulk mandates

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → digital payment does not mean UPI alone
- UPI → instant retail payment instruction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → digital payment does not mean UPI alone.

STAGE 06 OPERATOR REGULATOR PARTICIPANT

SEPARATE AADHAAR AUTHORITY

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• NPCI → operates specified retail rails • RBI → regulates payment systems	• BANK / PPI ISSUER → holds or issues funding instrument • APP PROVIDER → customer interface	• UIDAI → separate Aadhaar authority • NPCI → operates specified retail rails

06

STAGE 06 OPERATOR REGULATOR PARTICIPANT

OPERATOR REGULATOR PARTICIPANT OPERATES SPECIFIED RETAIL RAILS REGULATES PAYMENT SYSTEMS PPI ISSUER HOLDS OR ISSUES FUNDING INSTRUMENT APP PROVIDER CUSTOMER INTERFACE SEPARATE AADHAAR AUTHORITY

NPCI → operates specified retail rails

RBI → regulates payment systems

BANK / PPI ISSUER → holds or issues funding instrument

APP PROVIDER → customer interface

CONCEPT / ARCHITECTURE

- NPCI → operates specified retail rails
- RBI → regulates payment systems

MECHANISM / APPLICATION

- BANK / PPI ISSUER → holds or issues funding instrument
- APP PROVIDER → customer interface

READINESS / STATUS LIMIT

- UIDAI → separate Aadhaar authority
- NPCI → operates specified retail rails

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BANK / PPI ISSUER → holds or issues funding instrument.

07

STAGE 07 ACCOUNT AGGREGATOR FLOW

ACCOUNT AGGREGATOR FLOW GIVES EXPLICIT CONSENT HOLDS REGULATED FINANCIAL INFORMATION RETRIEVES AND TRANSFERS AS A DATA BLIND RECEIVES FOR CONSENTED PURPOSE AA DOES NOT OWN OR RETAIN THE INFORMATION

CUSTOMER → gives explicit consent

FIP → holds regulated financial information

AA → retrieves and transfers as a data blind

FIU → receives for consented purpose

RULE → AA does not own or retain the information

STARTING CONDITION / INPUT

- CUSTOMER → gives explicit consent
- FIP → holds regulated financial information

SCIENTIFIC OR ENGINEERING MECHANISM

- AA → retrieves and transfers as a data blind
- FIU → receives for consented purpose

OUTPUT / FAILURE BOUNDARY

- RULE → AA does not own or retain the information
- CUSTOMER → gives explicit consent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → AA does not own or retain the information.

08

STAGE 08 TRUST LAYERS

TRUST LAYERS ACCESS, SHARE AND VERIFY TRUSTED DOCUMENTS ESIGN ONLINE ELECTRONIC SIGNATURE UNDER TRUST FRAMEWORK IDENTITY AND AUTHENTICATION PAYMENT INSTRUCTION RAIL DOCUMENT, SIGNATURE, IDENTITY AND PAYMENT STAY SEPARATE

SYSTEM / DEVICE / INSTITUTION

DIGILOCKER → access, share and verify trusted documents eSIGN → online electronic signature under trust framework
AADHAAR → identity and authentication

DECISIVE TECHNICAL DISTINCTION

UPI → payment instruction rail
RULE → document, signature, identity and payment stay separate

STATUS / UNIT / EVIDENCE LIMIT

DIGILOCKER → access, share and verify trusted documents eSIGN → online electronic signature under trust framework
AADHAAR → identity and authentication

SYSTEM / DEVICE / INSTITUTION

- DIGILOCKER → access, share and verify trusted documents eSIGN → online electronic signature under trust framework
- AADHAAR → identity and authentication

DECISIVE TECHNICAL DISTINCTION

- UPI → payment instruction rail
- RULE → document, signature, identity and payment stay separate

STATUS / UNIT / EVIDENCE LIMIT

- DIGILOCKER → access, share and verify trusted documents eSIGN → online electronic signature under trust framework
- AADHAAR → identity and authentication

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → document, signature, identity and payment stay separate.

09

STAGE 09 OPEN-NETWORK EXTENSIONS

OPEN-NETWORK EXTENSIONS OPEN COMMERCE PROTOCOL OPEN CREDIT PROTOCOL NETWORK LOGIC UNBUNDLE FRONT ENDS FROM SHARED RULES COMPETE ON TOP OF PROTOCOL PROTOCOL IS NOT A PRIVATE MARKETPLACE OR LENDER

ONDC → open commerce protocol

OCEN → open credit protocol

NETWORK LOGIC → unbundle front ends from shared rules

APPLICATIONS → compete on top of protocol

CONCEPT / ARCHITECTURE

- ONDC → open commerce protocol
- OCEN → open credit protocol

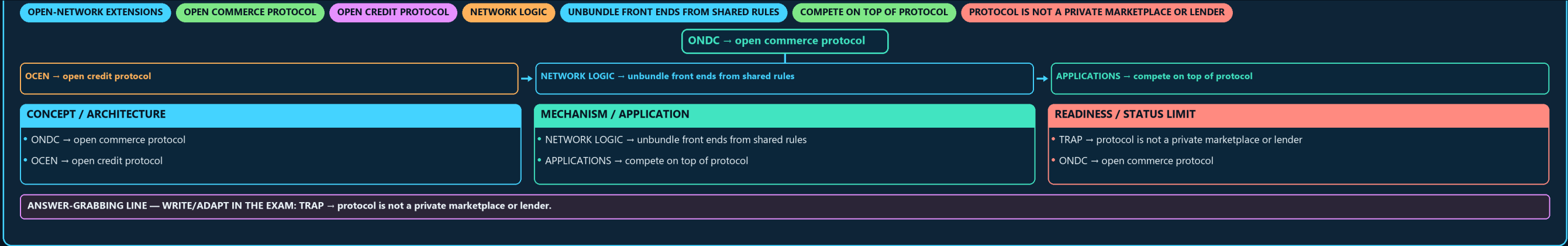
MECHANISM / APPLICATION

- NETWORK LOGIC → unbundle front ends from shared rules
- APPLICATIONS → compete on top of protocol

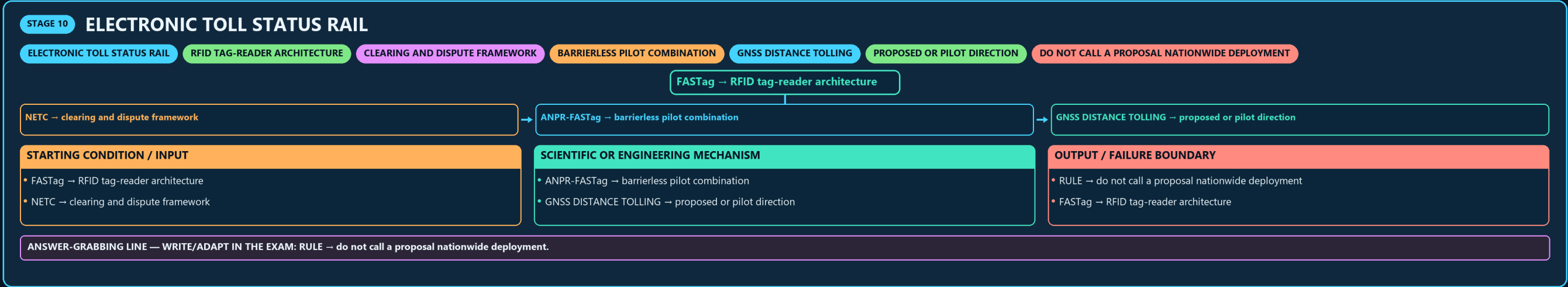
READINESS / STATUS LIMIT

- TRAP → protocol is not a private marketplace or lender
- ONDC → open commerce protocol

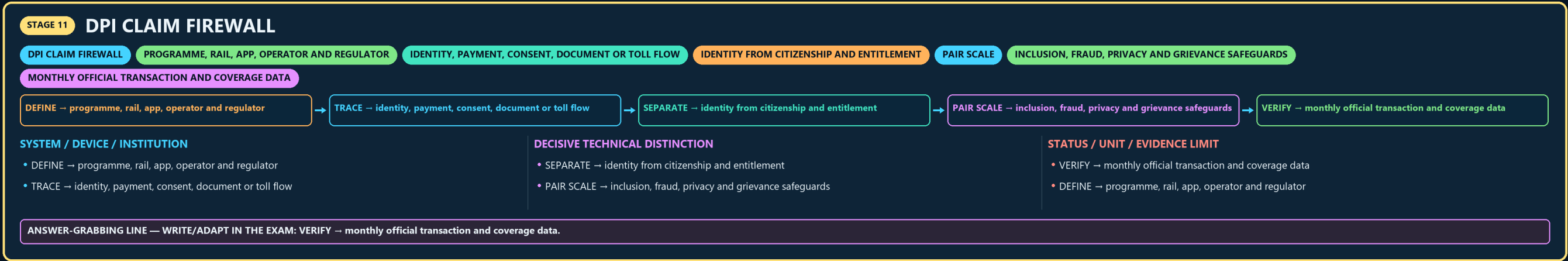
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → protocol is not a private marketplace or lender.



10



11



E

